

System of Linear Equations

$$\begin{cases} x + 2y + z = 5 \\ 2x + 4y + 2z = 10 \\ x + 3y - z = 4 \end{cases}$$

Augmented Matrix $[A : B]$

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 2 & 4 & 2 & 10 \\ 1 & 3 & -1 & 4 \end{array} \right]$$

Row Reduction

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 2 & 4 & 2 & 10 \\ 1 & 3 & -1 & 4 \end{array} \right] \quad R_2 \leftarrow R_2 - 2R_1 \Rightarrow R_2 = (0, 0, 0, 0)$$
$$R_3 \leftarrow R_3 - R_1 \Rightarrow R_3 = (0, 1, -2, -1)$$

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & -1 \end{array} \right] \quad \text{Swap } R_2 \leftrightarrow R_3$$

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad R_1 \leftarrow R_1 - 2R_2 \Rightarrow R_1 = (1, 0, 5, 7)$$

$$[A : B] = \left[\begin{array}{ccc|c} 1 & 0 & 5 & 7 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

General and Particular Solution

General Solution

Let $z = t$, $t \in \mathbb{R}$

$$\begin{cases} x = 7 - 5t \\ y = -1 + 2t \\ z = t \end{cases}$$

Particular Solution ($t = 0$)

$$(x, y, z) = (7, -1, 0)$$